Quantification: 682_z23
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	14
interior lacuna count	11
frame-edge lacuna count	3

Interior lacunae only: mean, median and sample SD (n = 11)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	1994.91	1519	1026.72	px^2
major_axis_length_px	lacuna shape	85.7455	86.1566	22.5145	px
minor_axis_length_px	lacuna shape	30.4546	26.8577	8.8339	px
aspect_ratio	lacuna shape	2.9668	3.0363	0.8704	unitless
eccentricity	lacuna shape	0.92	0.9442	0.0643	unitless
solidity	lacuna shape	0.8555	0.8762	0.0782	unitless
roots_count	headline	8.6364	9	2.6181	count
ring_length_r30_px	headline	348.364	352	87.1324	px
ring_length_r60_px	reported	900.909	900	167.645	px
owned_length_px	ownership-dependent	2403.53	1927.37	1454.46	px
edge_count	ownership-dependent	112	96	71.8999	count
mean_edge_length_px	ownership-dependent	22.1506	21.3106	3.1481	px
perimeter_px	normalised	221.572	210.244	51.1926	px
ring_area_r30_px2	normalised	8909.82	8646	1432.67	px^2
ring_area_r60_px2	normalised	22488.8	22056	3166.3	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	0.9972	1	0.0094	unitless
ring_density_r30	normalised	0.0396544	0.0372427	0.0115157	px^-1
ring_density_r60	normalised	0.0404533	0.0396142	0.00805999	px^-1
roots_per_100px_perimeter	normalised	4.0142	3.6416	1.4325	per 100 px
ring_attached_length_r30_px	network v2	302.273	292	101.17	px
ring_attached_length_r60_px	network v2	674.455	605	230.608	px
ring_length_w_r30_px	network v2	393.825	392.735	98.2953	px
ring_length_w_r60_px	network v2	1017.43	1012.57	185.546	px
sholl_crossings_r10	network v2	8.3636	8	1.9633	count
sholl_crossings_r20	network v2	11.0909	12	2.7002	count
sholl_crossings_r30	network v2	12.8182	12	3.5726	count
ring_width_mean_r30_px	width	7.1663	7.2028	0.3366	px

Per field
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measure	value	units
analysed_area_px2	1.02271e+06	px^2
total_skeleton_length_px	38363	px
canalicular_length_density_per_px	0.037511	px^-1
junction_count	880	count
junction_density_per_px2	0.00086046	px^-2
lacuna_count	14	count
lacunae_per_px2	1.369e-05	px^-2
skeleton_component_count	263	count

mean_component_length_px	145.867	px
median_component_length_px	17	px
field_length_density_w_per_px	0.0424887	px^-1
field_density_without_flagged_per_px	0.0385419	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	560	px
unattached_component_count	30	count
canalicular_length_density_attached_per_px	0.037511	px^-1
field_length_density_with_unattached_per_px	0.0380586	px^-1

Canalicular width, the new measure

measure	value	units
width_mean_px	7.7464	px
width_median_px	8	px
width_p10_px	4.4721	px
width_p90_px	10.198	px
width_pixels_used	36990	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	510	36.0388	20.5481	1.7539	0.8215	0.8095	-0.2657	13.3863	106.873	True
2	448	43.7126	19.8458	2.2026	0.891	0.6767	1.435	7.0045	363.067	True
3	1379	89.4664	21.3431	4.1918	0.9711	0.7971	-1.5436	60.8564	471.593	False
4	1062	83.8665	19.2697	4.3522	0.9732	0.6923	-1.1353	135.124	920.395	False
5	4132	136.42	39.4244	3.4603	0.9573	0.9008	1.5344	431.979	214.138	False
6	1032	65.9485	21.7201	3.0363	0.9442	0.8418	-0.9507	441.636	703.582	False
7	2868	91.0398	40.6809	2.2379	0.8946	0.9166	-1.49	534.719	477.113	False
8	980	60.8458	24.7024	2.4632	0.9139	0.7644	-1.4596	557.819	348.526	False
9	2962	111.965	35.5084	3.1532	0.9484	0.917	1.4275	591.514	47.6516	True
10	2810	108.692	33.8251	3.2134	0.9503	0.9201	1.4475	714.117	172.09	False
11	1297	66.0375	26.8577	2.4588	0.9136	0.9134	-1.189	725.743	851.213	False
12	1519	86.1566	25.1812	3.4215	0.9563	0.8388	1.3918	837.228	113.164	False
13	2083	63.4325	42.3049	1.4994	0.7451	0.9485	-1.2565	905.171	752.206	False
14	2782	91.2948	39.6911	2.3001	0.9005	0.8762	-1.4405	919.61	300.233	False

Network per lacuna

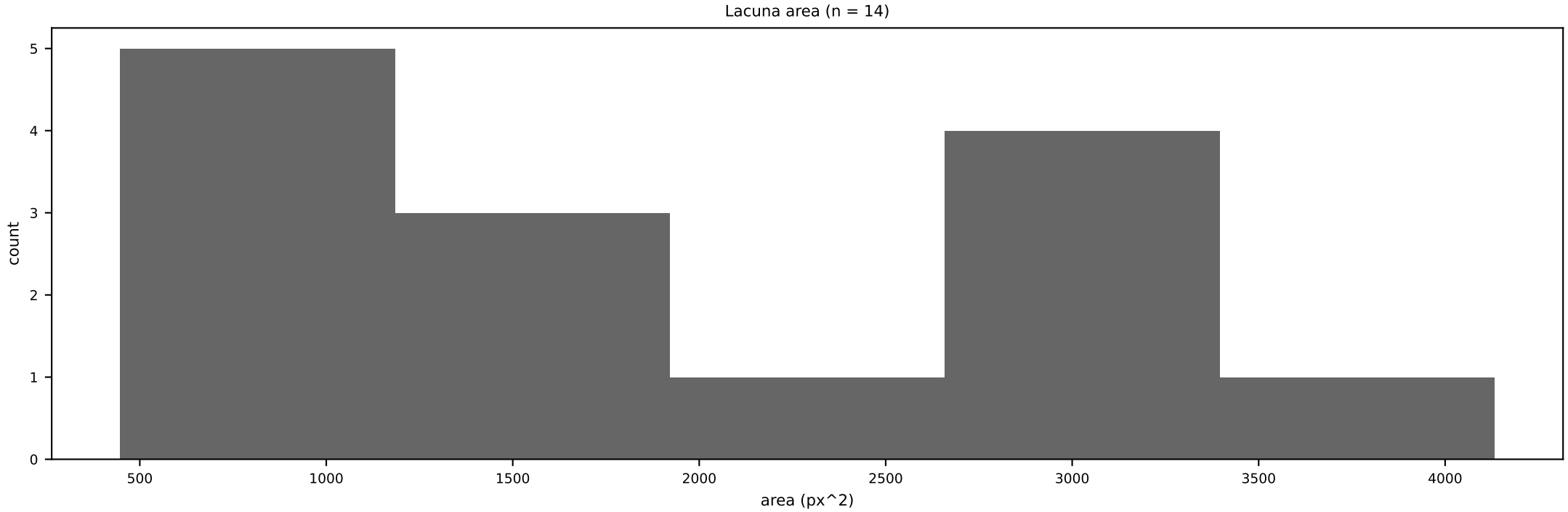
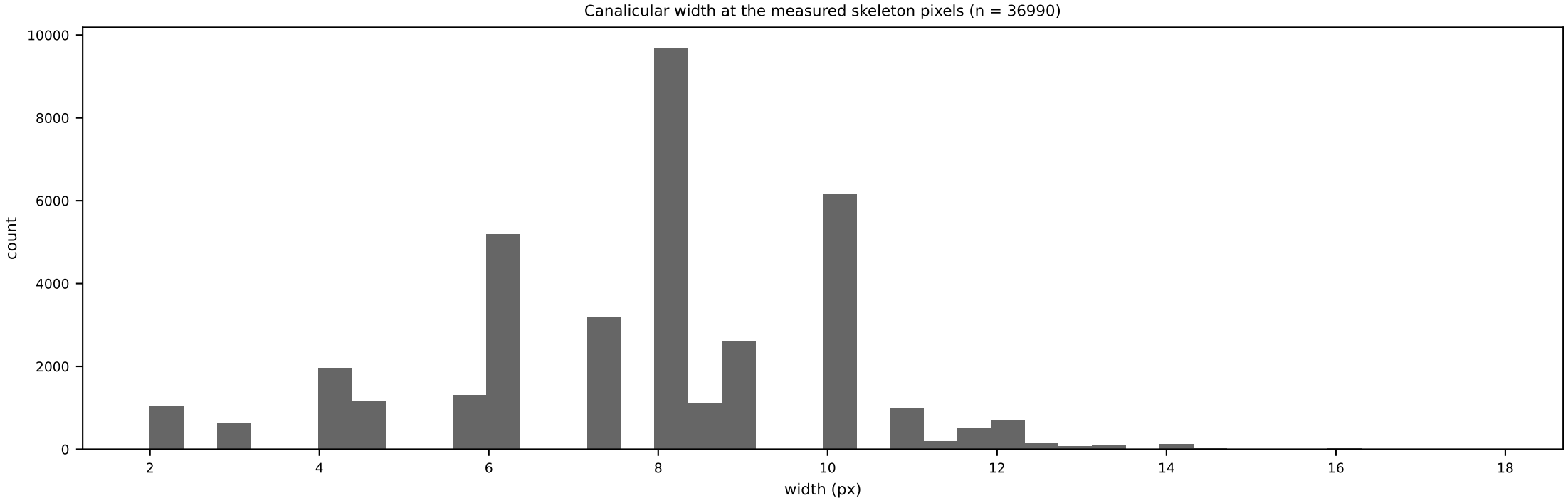
lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	2	96	220	3079.49	141	21.8404	
2	3	42	91	36.3137	4	9.0784	
3	8	329	752	1927.37	96	20.0767	6.7412
4	12	461	1086	5431.62	268	20.2673	7.2028
5	9	352	922	3128.17	156	20.0524	7.4414
6	7	261	743	1344.83	62	21.6908	7.6091
7	7	369	900	3942.46	185	21.3106	7.2722
8	12	490	1003	2452.03	116	21.1382	7.4172
9	10	326	785	1237.61	54	22.9187	7.5147
10	9	398	1191	1534.31	67	22.9002	7.5386

11	6	249	659	1140.41	37	30.822	6.6908
12	9	322	879	1209.22	61	19.8232	6.6855
13	4	208	737	776.784	36	21.5773	7.1602
14	12	393	1038	3551.65	148	23.9976	7.0702

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	105.154	3701	10161	0.6475	0.597	0.0259389	0.0216514	1.902
2	132.847	3699	7686	0.6031	0.5656	0.0113544	0.0118397	2.2582
3	219.681	8976	20715	1	0.9689	0.0366533	0.0363022	3.6416
4	210.244	8511	22591	1	1	0.0541652	0.0480722	5.7077
5	335.421	12178	29932	1	1	0.0289046	0.0308032	2.6832
6	167.238	7485	20582	1	1	0.0348697	0.0360995	4.1857
7	243.522	9472	22056	1	1	0.0389569	0.0408052	2.8745
8	167.924	7349	17555	1	1	0.0666757	0.0571347	7.1461
9	282.894	8678	19332	0.813	0.7462	0.0375663	0.0406063	3.5349
10	263.48	10172	24552	1	1	0.039127	0.0485093	3.4158
11	174.51	7570	20770	1	1	0.032893	0.0317284	3.4382
12	210.167	8646	22189	1	1	0.0372427	0.0396142	4.2823
13	185.823	7980	21610	1	1	0.0260652	0.0341046	2.1526
14	259.279	9669	24825	1	1	0.0406454	0.0418127	4.6282

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	81	151	103.426	241.823	3	2	4
2	38	38	43.1421	98.7696	1	0	0
3	190	462	366.907	831.955	7	12	14
4	450	1055	531.517	1247.16	11	13	12
5	325	605	392.735	1022.42	8	12	9
6	224	403	301.836	856.737	7	6	7
7	292	602	411.149	1012.57	9	13	12
8	478	976	552.818	1123.74	11	13	15
9	271	619	366.149	879.169	8	11	11
10	345	889	448.12	1336.56	8	13	19
11	190	412	282.865	763.696	5	10	10
12	262	623	370.848	991.051	8	9	17
13	202	540	235.238	840.767	7	7	11
14	367	852	438.049	1165.06	11	14	15

682_z23: distributions, pre-validation, px